

Mesa Card + LinuxCNC Setup Guide

English translation of the vendor manual (Mesa 7192/7198 Ethernet FPGA card). All screenshots and diagrams have been translated. The linked tutorial videos are in Chinese.

1. Change the host computer's IP address

The host must join the Mesa card's subnet before LinuxCNC can talk to the card. The card ships at address **10.10.10.10**. Configure the wired connection manually with two IP rows so the host can reach both your LAN and the card.



Fig. 1 – Right-click the network icon in the system tray, then click “Edit Connections...”.

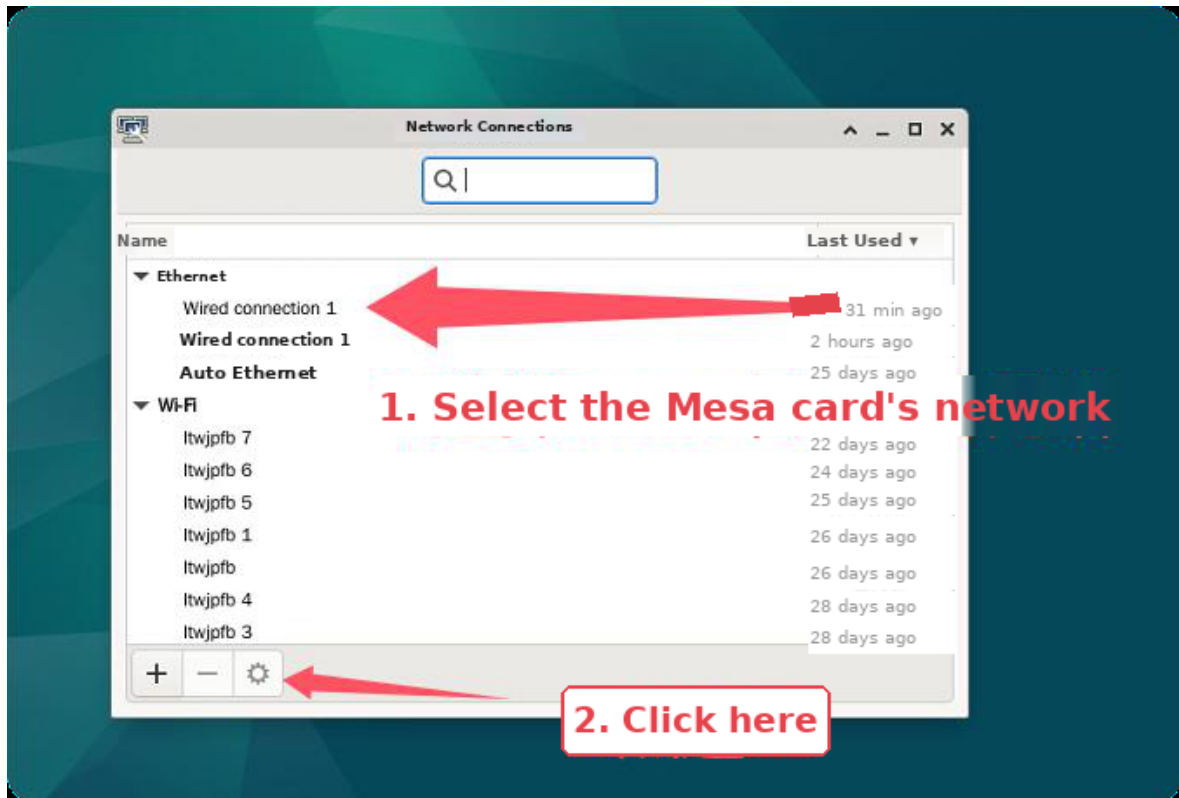


Fig. 2 – Select the network the Mesa card is connected to, then click the gear button.

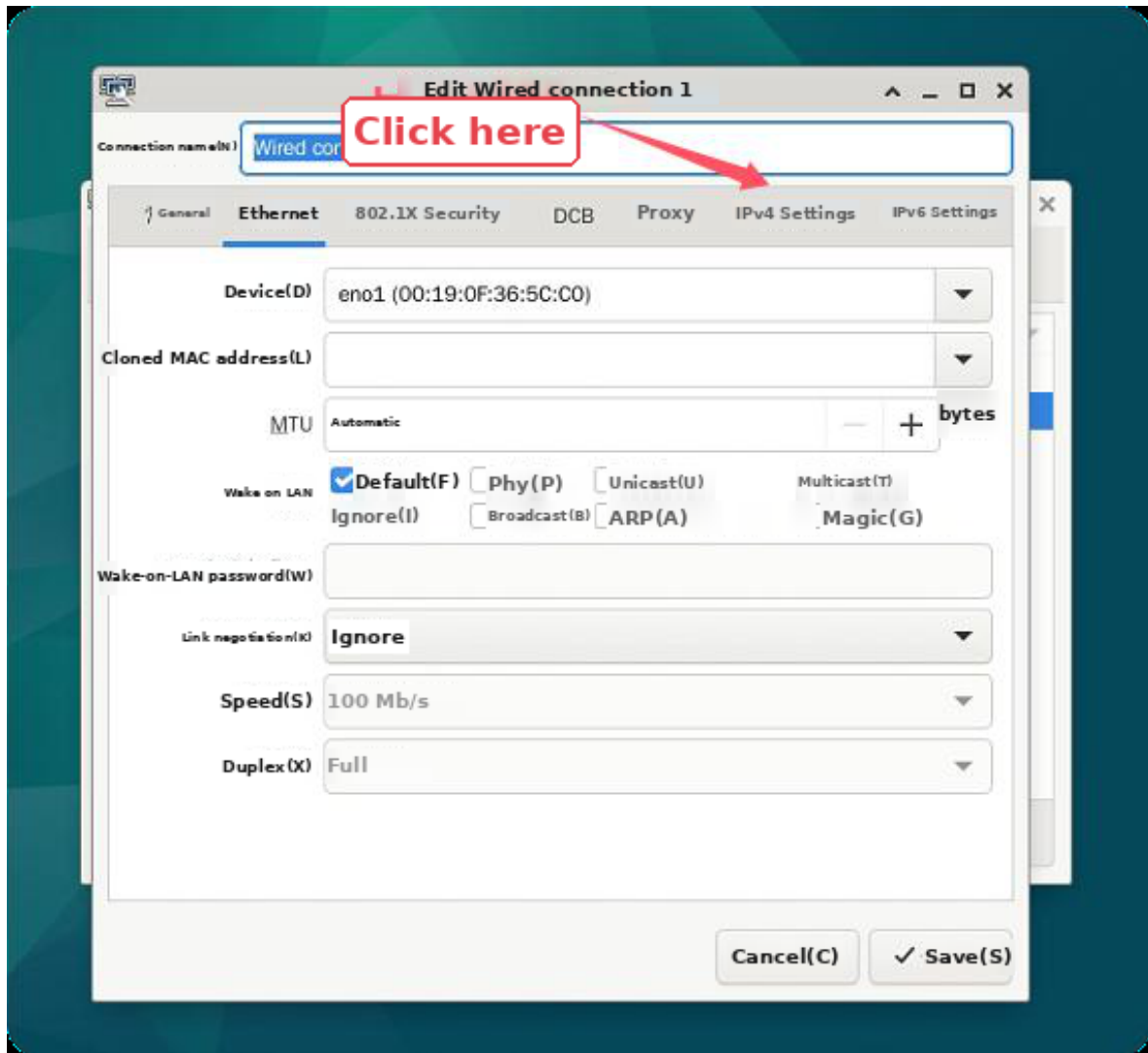


Fig. 3 – In the connection editor, open the "IPv4 Settings" tab.

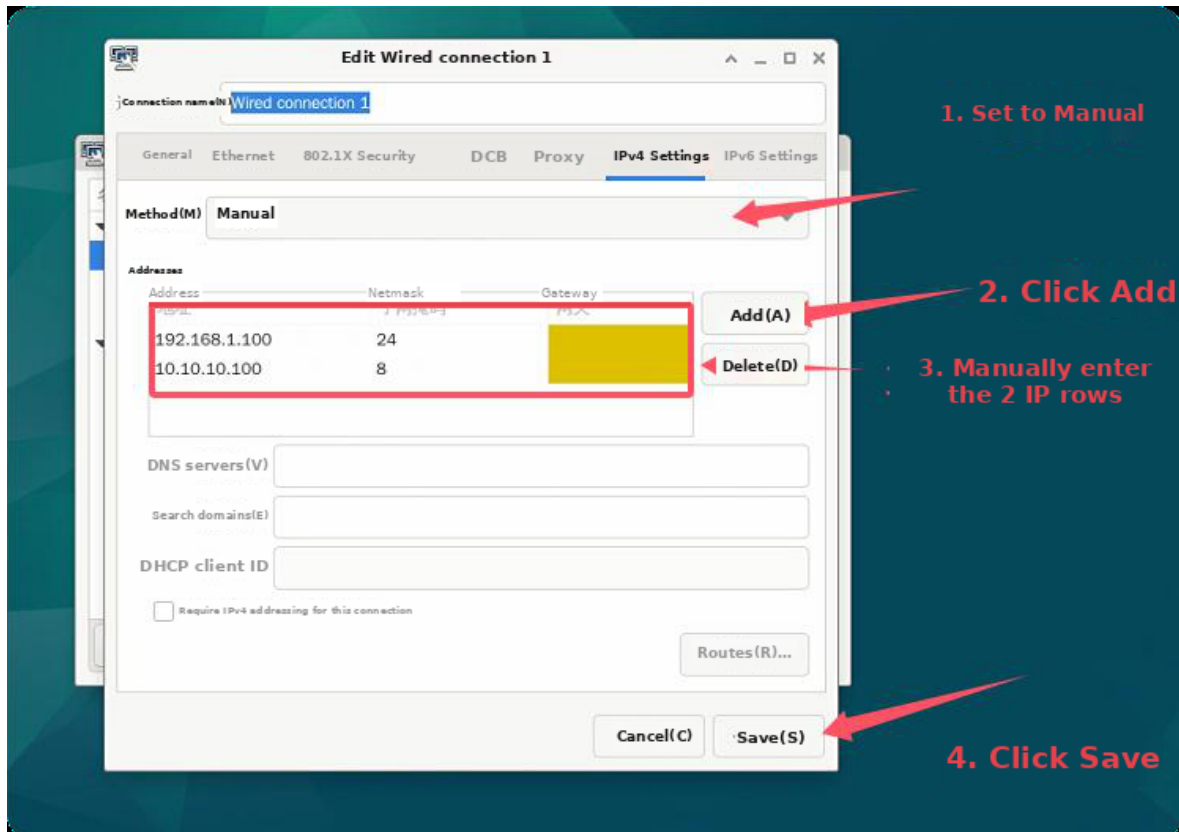


Fig. 4 – Set Method to “Manual”, click “Add” and enter the two rows below, then click Save.

Address	Netmask	(Gateway: leave empty)
192.168.1.100	24	
10.10.10.100	8	

2. Optimize the host computer

Two optimizations are required for reliable real-time behavior:

Kernel isolation (mandatory). Dedicate one CPU core exclusively to LinuxCNC (isolcpus). This is essential for stable servo-thread timing with an Ethernet Mesa card.

BIOS settings. Disable virtualization, disable hyper-threading (**must** be disabled), disable turbo boost, and disable power-saving states.

Video tutorial (Chinese): <https://www.bilibili.com/video/BV1cP8MzmE8F/>

3. Configure the Mesa card (pncconf wizard)

Run the **pncconf** wizard to detect and configure the card.

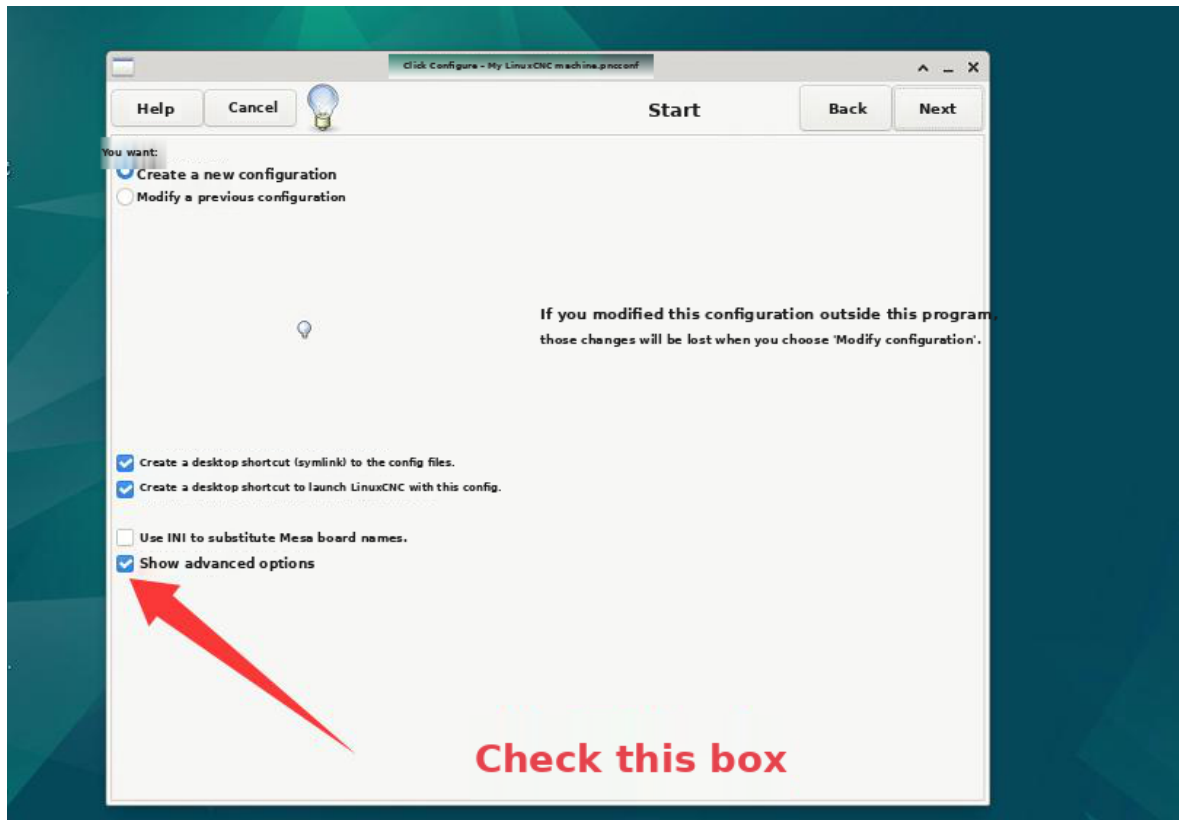


Fig. 5 – On the first page, tick “Show advanced options”.

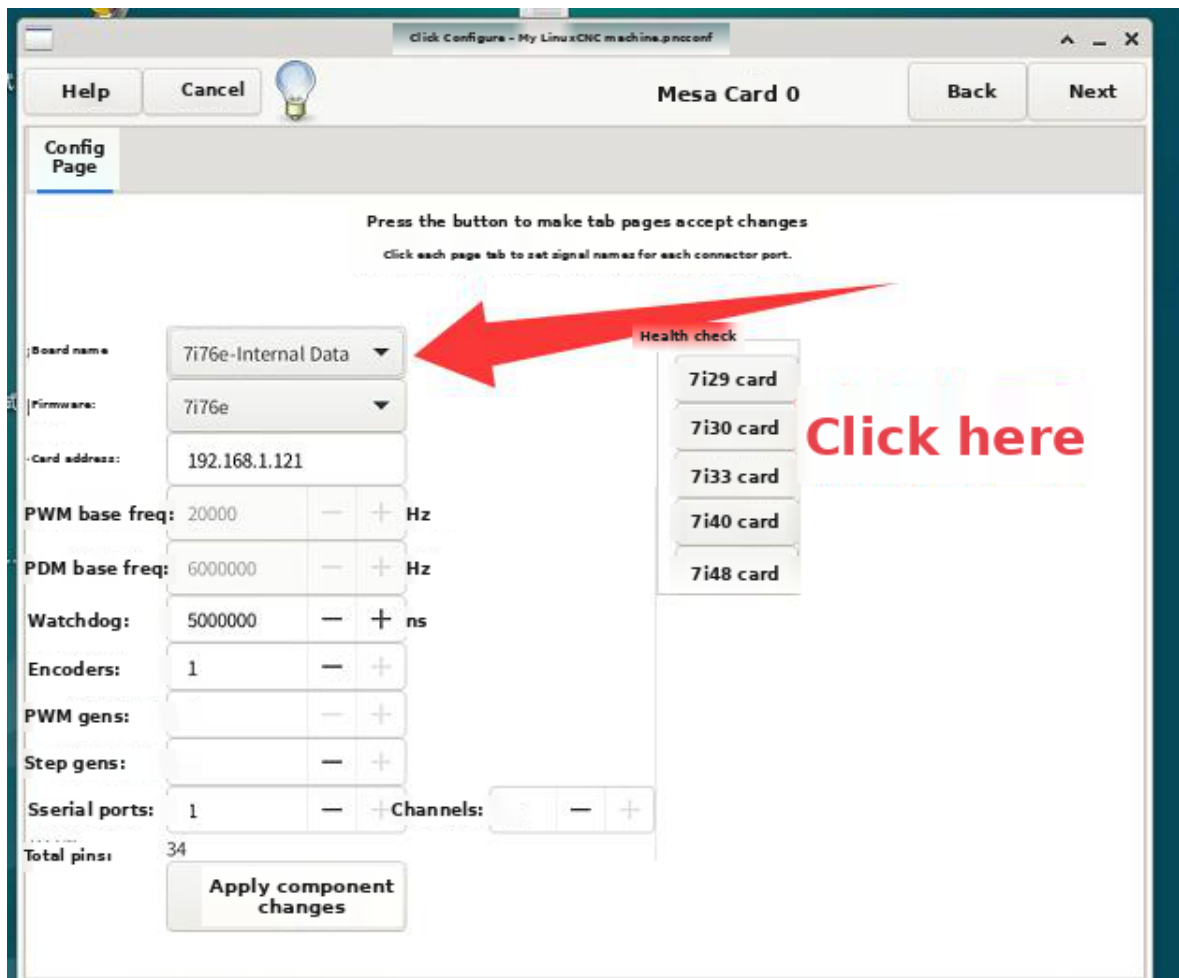


Fig. 6 – On the Mesa Card 0 page, open the “Board name” dropdown.

Click Configure - My LinuxCNC machine.pncconf

Help Cancel  Mesa Card 0 Back Next

Config Page

Press the button to make tab pages accept changes
Click each page tab to set signal names for each connector port.
Pinmux connector requires a valid pinmux table for the pinmux table.

Board name: Discovery Option ▾ Health check

Firmware: ▾

Card address: 192.168.1.121

PWM base freq: 20000 — + Hz

PDM base freq: 6000000 — + Hz

Watchdog: 5000000 — + ns

Encoders: 1 — +

PWM gens: — +

Step gens: — +

Serial ports: 1 — + Channels: 2 — +

Total pins: 34

Apply component changes Board Discovery

7i29 card
7i30 card
7i33 card
7i40 card
7i48 card

Select this

Fig. 7 – Select “Discovery Option”.

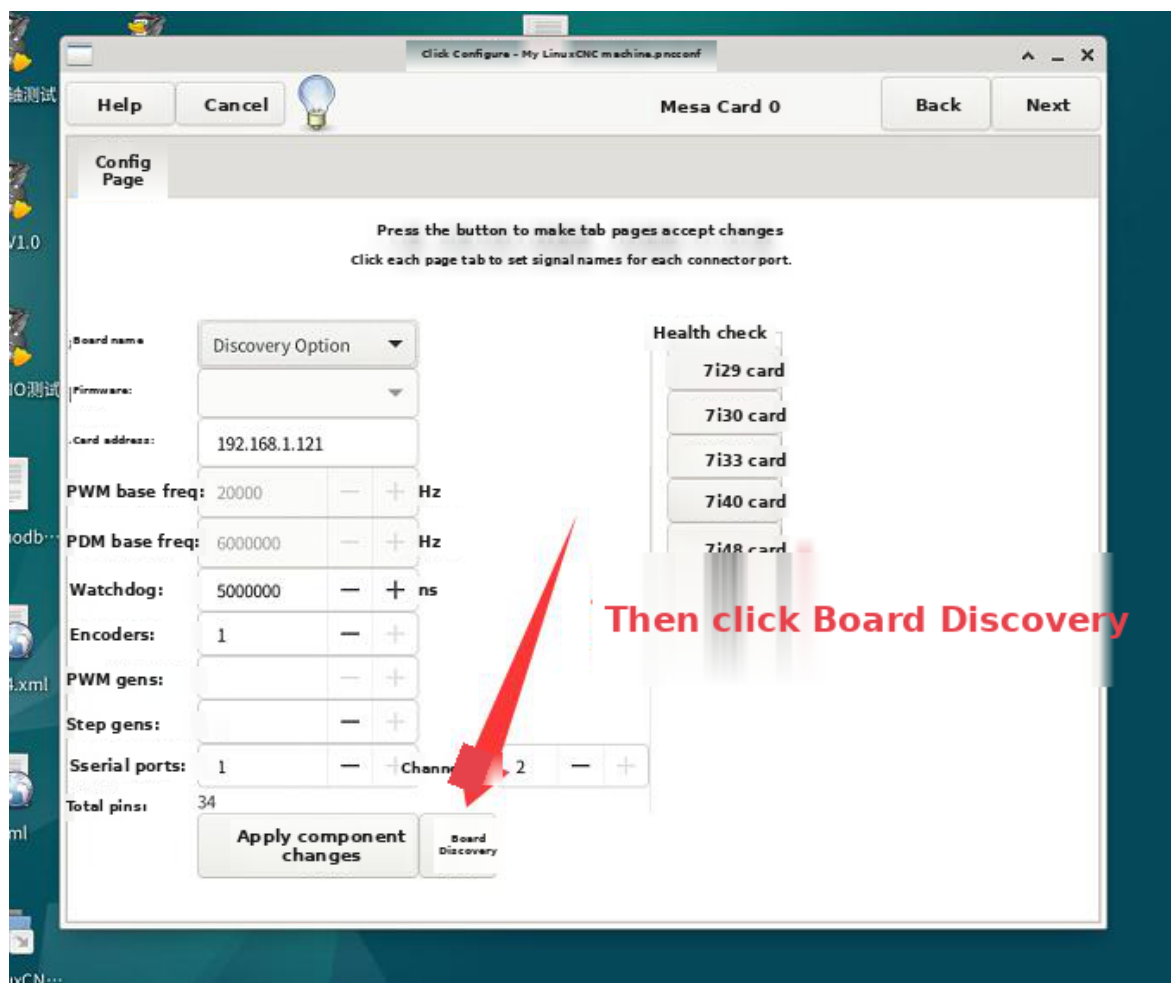


Fig. 8 – Then click “Board Discovery”.

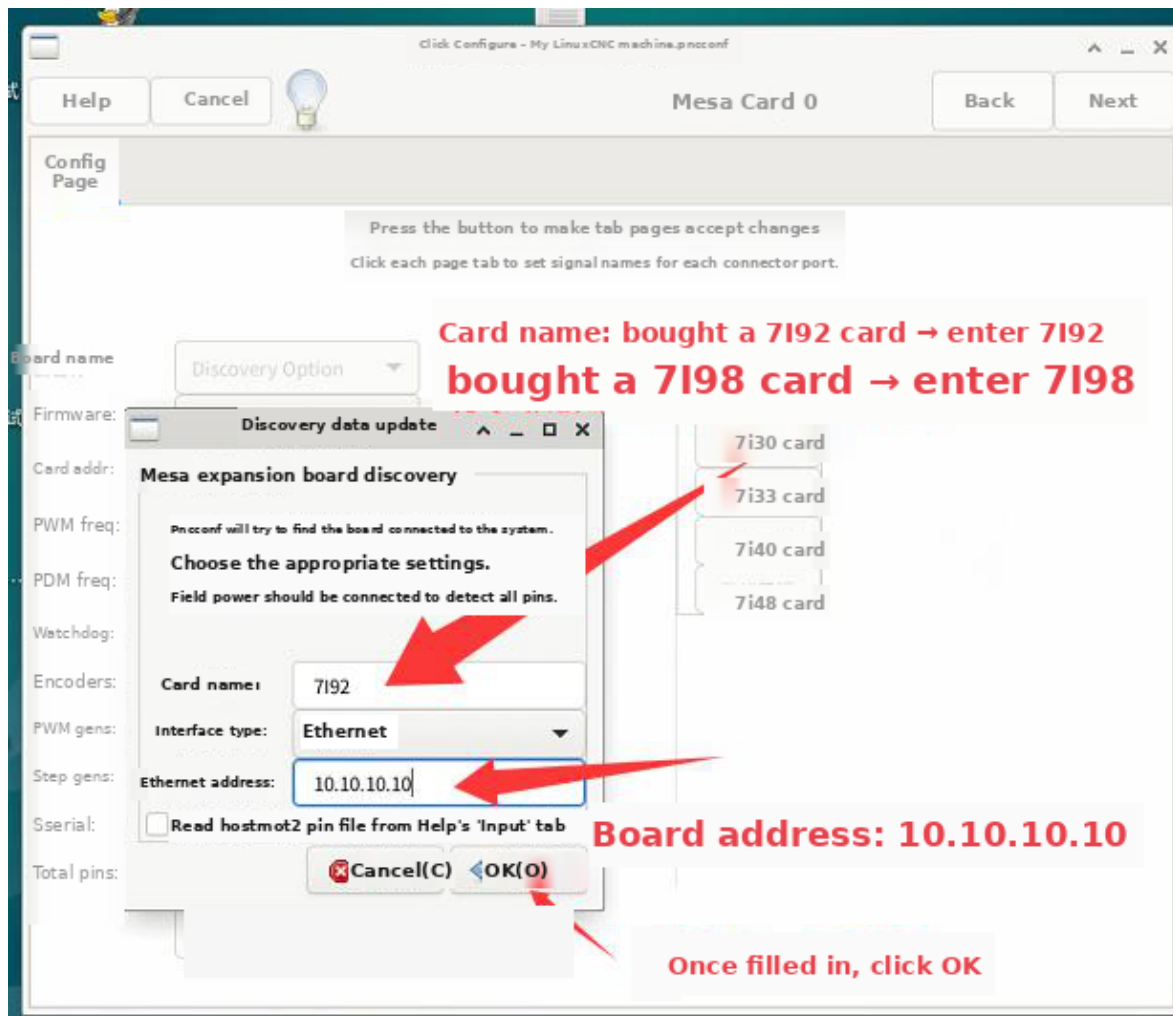


Fig. 9 – Enter the card name (7I92 if you bought a 7I92, 7I98 if you bought a 7I98), interface type “Ethernet”, address 10.10.10.10, then click OK.

Video tutorial (Chinese): <https://www.bilibili.com/video/BV17m7zzvE1Q/>

4. Fixing random disconnections (“error finishing read!”)

If gmoccapy shows the error below, the real-time link to the card is dropping packets:

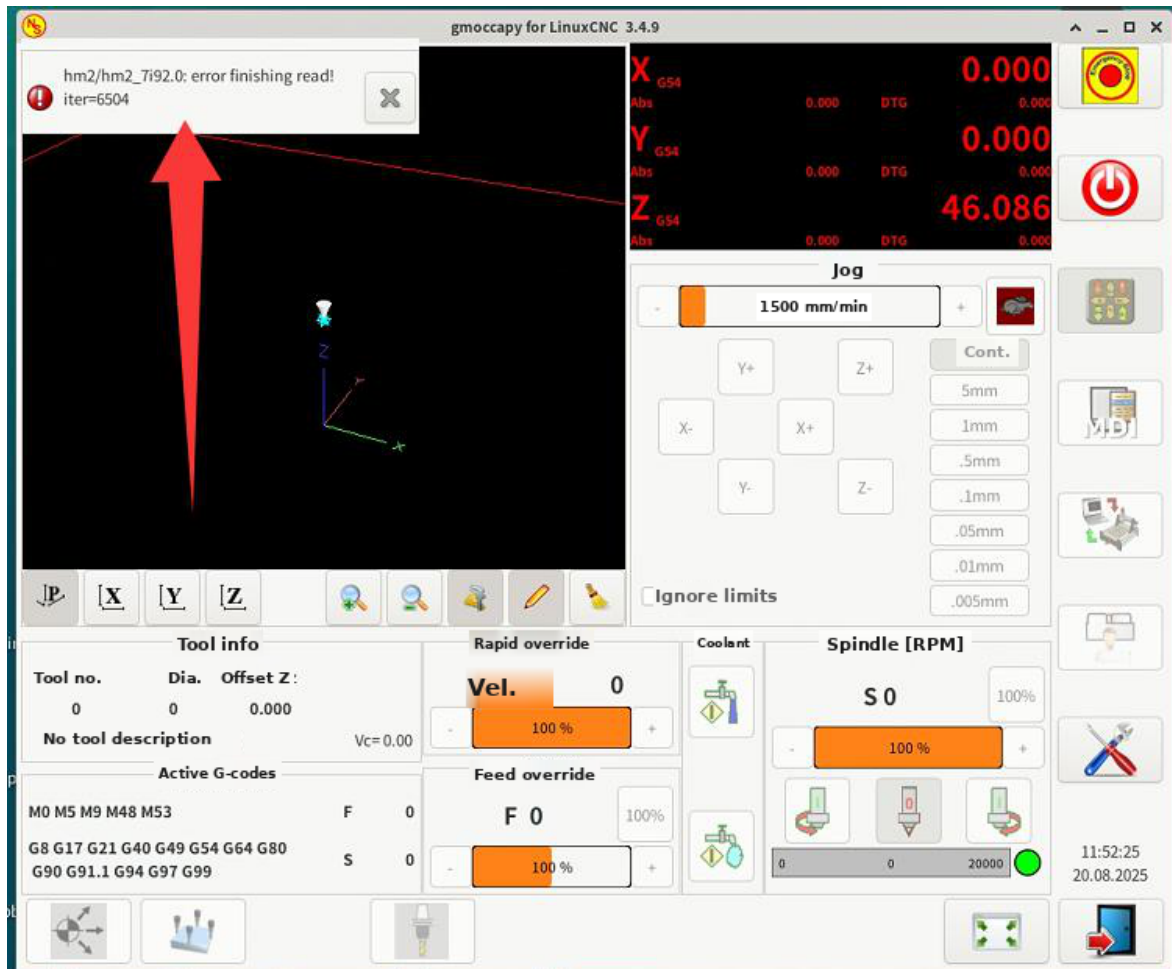


Fig. 10 – gmoccapy error: "hm2/hm2_7i92.0: error finishing read!"

Work through these steps in order:

Step 1. Make sure kernel isolation and the BIOS settings from Section 2 are done.

Step 2. Try a better shielded Ethernet cable.

Step 3. Bind the network card's interrupt (IRQ) to the isolated CPU core. First find the IRQ number of the NIC the Mesa card is connected to:

```
$ cat /proc/interrupts
```

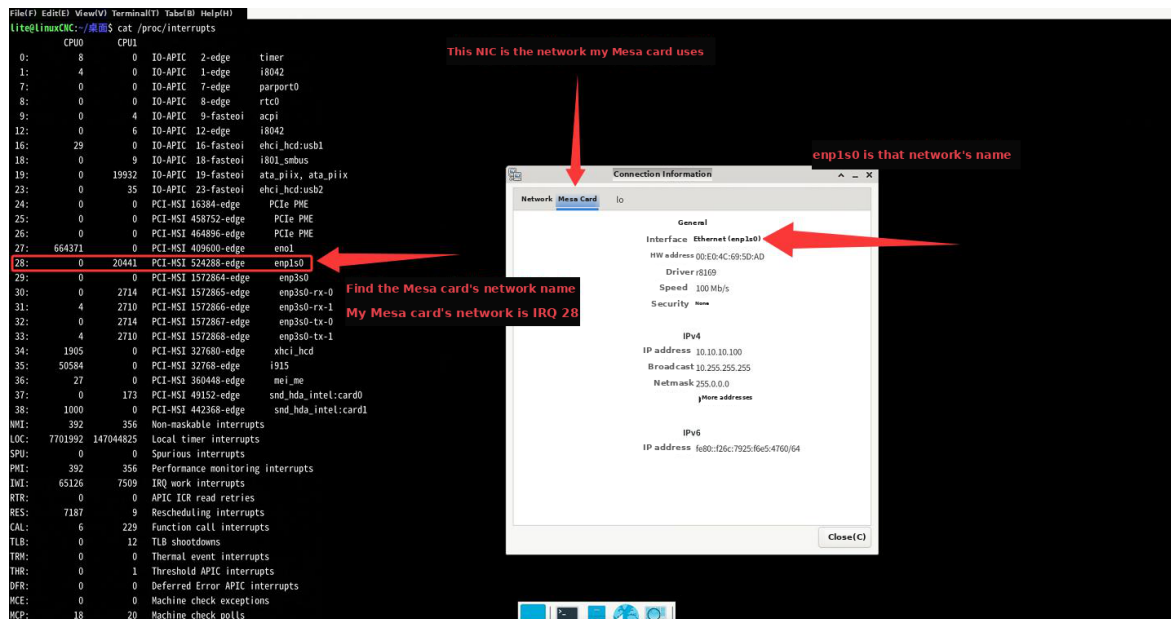


Fig. 11 – Find the NIC (here enp1s0). In this example its interrupt is IRQ 28. The Connection Information dialog confirms which interface has address 10.10.10.100.

Then bind that IRQ to the isolated core. The value written is a **bitmask**: CPU0=1, CPU1=2, CPU2=4, CPU3=8. On a 2-core machine with core 1 isolated, write 2; on a 4-core machine with core 3 isolated, write 8:

2-core machine, core 1 isolated:
echo 2 > /proc/irq/28/smp_affinity

4-core machine, core 3 isolated:
echo 8 > /proc/irq/28/smp_affinity

```
bash
echo 8 > /proc/irq/28/smp_affinity
```

binds IRQ 28 to CPU3.

Because smp_affinity's value is a bitmask, mapping to CPUs in binary:

CPU no.	Mask (decimal)	Mask (binary)
CPU0	1	0001
CPU1	2	0010
CPU2	4	0100
CPU3	8	1000

Fig. 12 – smp_affinity is a bitmask: the value maps to CPUs in binary.

Verify with cat /proc/irq/28/smp_affinity. To make the binding permanent, create /etc/rc.local:

```
#!/bin/bash
sleep 3
```

```
echo 2 > /proc/irq/28/smp_affinity
exit 0

sudo chmod +x /etc/rc.local
sudo systemctl enable rc-local
sudo systemctl start rc-local
```

5. Wiring diagrams

Driver common-anode wiring

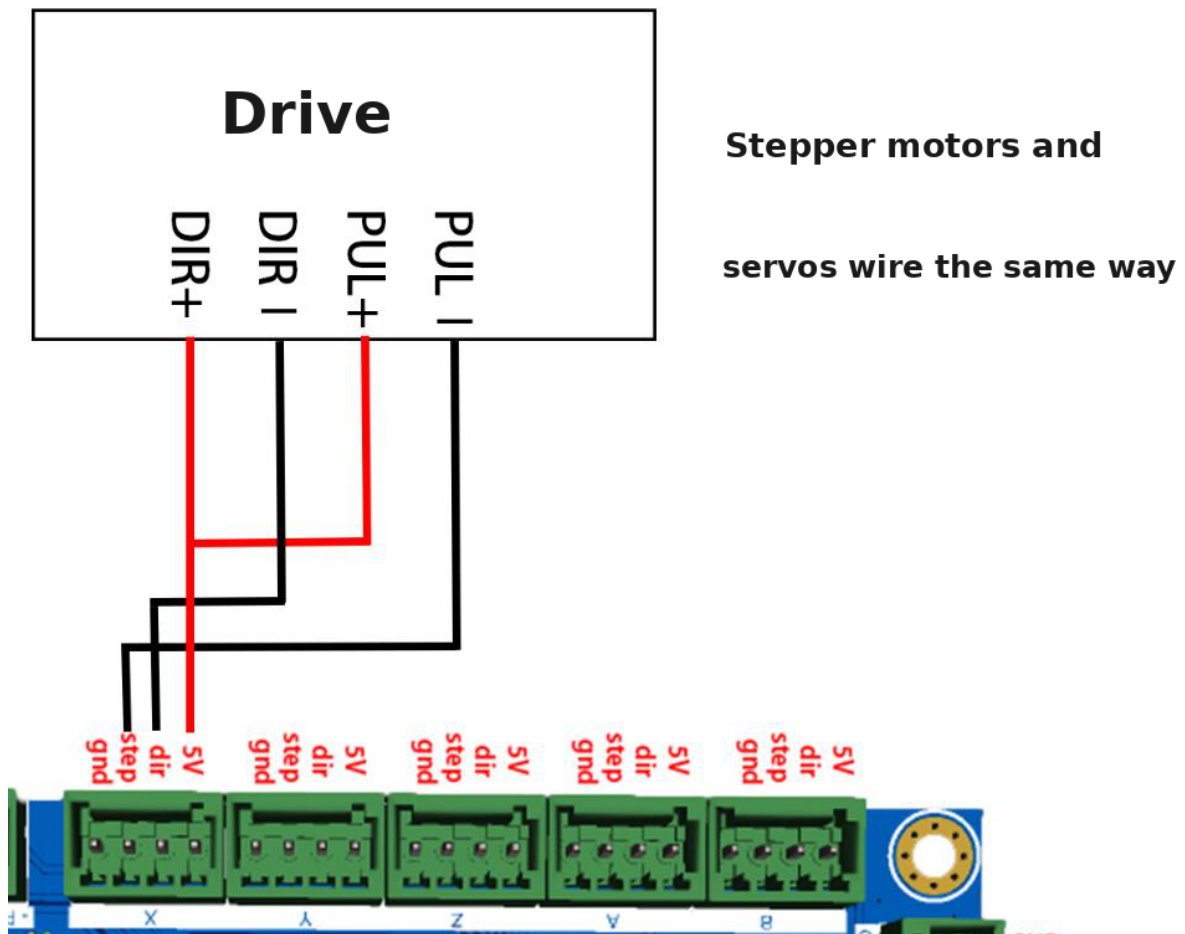


Fig. 13 – Stepper/servo driver, common-anode wiring: DIR+ and PUL+ to 5V; DIR- to dir, PUL- to step. Stepper and servo drives wire the same way.

Mechanical switch wiring

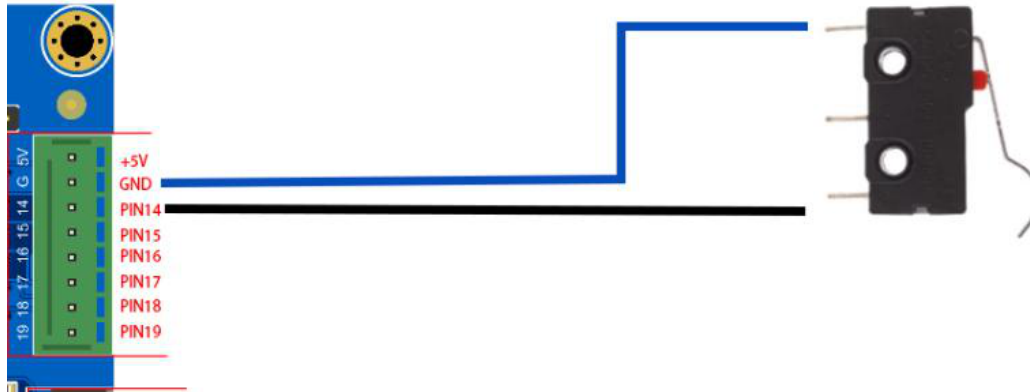


Fig. 14 – Mechanical switch wiring: between GND and PIN14.

Proximity switch & slot photo-sensor wiring

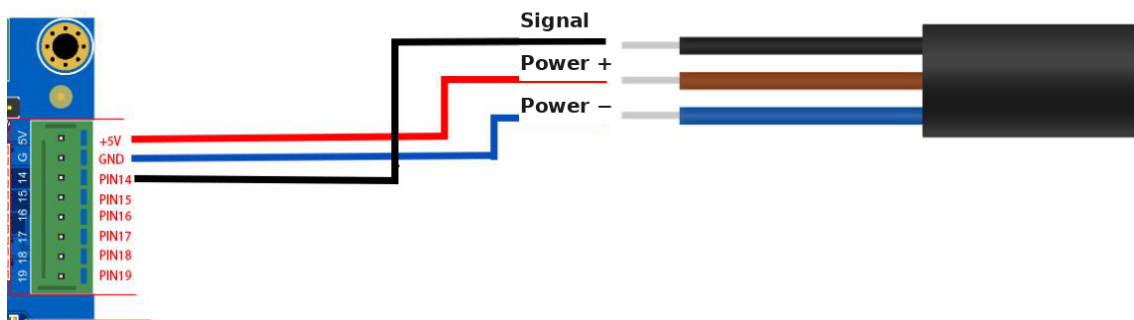


Fig. 15 – Proximity switch / slot photo-sensor wiring: signal (black) to PIN14, power + (brown) to +5V, power – (blue) to GND.

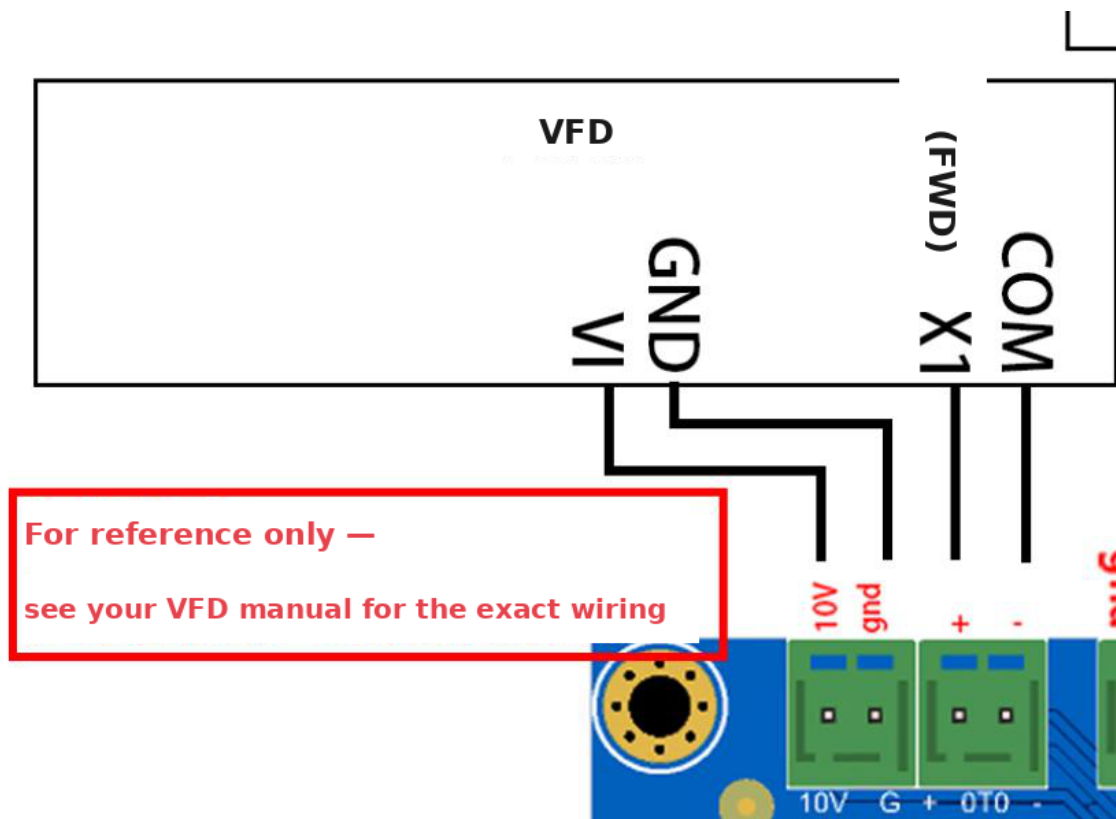


Fig. 16 – VFD (inverter) wiring: VI/GND to 10V/G, X1 (forward run) and COM to + / – 0T0. For reference only — check your VFD manual for the exact wiring.